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Equatorial origin for Lower Jurassic radiolarian chert in the Franciscan Complex, San Rafael Mountains, southern California

Jonathan T. Hagstrum, Benita L. Murchey, and Robert S. Bogar[1]
U.S. Geological Survey, Menlo Park, California

Abstract. Lower Jurassic radiolarian chert sampled at two localities in the San Rafael Mountains of southern California (~20 km north of Santa Barbara) contains four components of remanent magnetization. Components A, B', and B are inferred to represent uplift, Miocene volcanism, and subduction/accretion overprint magnetizations, respectively. The fourth component (C), isolated between 580° and 680°C, shows a magnetic polarity stratigraphy and is interpreted as a primary magnetization acquired by the chert during, or soon after, deposition. Both sequences are late Pliensbachian to middle Toarcian in age, and an average paleolatitude calculated from all tilt-corrected C components is 1° ± 3° north or south. This result is consistent with deposition of the cherts beneath the equatorial zone of high biologic productivity and is similar to initial paleolatitudes determined for chert blocks in northern California and Mexico. This result supports our model in which deep-water Franciscan-type cherts were deposited on the Farallon plate as it moved eastward beneath the equatorial productivity high, were accreted to the continental margin at low paleolatitudes, and were subsequently distributed northward by strike-slip faulting associated with movements of the Kula, Farallon, and Pacific plates. Upper Cretaceous turbidites of the Cachuma Formation were sampled at Agua Caliente Canyon to determine a constraining paleolatitude for accretion of the Jurassic chert sequences. These apparently unaltered rocks, however, were found to be completely overprinted by the A component of magnetization. Similar in situ directions and demagnetization behaviors observed in samples of other Upper Cretaceous turbidite sequences in southern and Baja California imply that these rocks might also give unreliable results.

Introduction

Blocks of Jurassic deep-marine radiolarian chert, offscraped from consumed oceanic crust, are found within subduction complexes along the western American margin from Central America to Alaska. Paleomagnetic data for relatively unaltered red chert in the Franciscan Complex of northern California indicate that these rocks were deposited within the equatorial zone of high biologic productivity [*Hagstrum and Murchey*, 1993]. Sedimentologic [*Karl*, 1984] and biostratigraphic [*Murchey*, 1984] evidence indicating gradual movement of the depositional site away from peak equatorial productivity is also consistent with the paleomagnetic data. Moreover, plate motion models [*Engebretson et al.*, 1985] corroborate low-latitude trajectories (e.g., 3°S to 11°N) of the Farallon plate from ridge crest to subduction zone, implying accretion of the northern California chert sequences at low paleolatitudes during the Cenomanian [*Hagstrum and Murchey*, 1993].

Paleomagnetic data for Franciscan-type red chert in western Baja California also indicate that these rocks were deposited beneath the equatorial high-productivity zone [*Hagstrum and Sedlock*, 1992]. Reliable paleomagnetic data are presently unavailable from chert sequences in Central America, but radiolarian faunas within the Nicoya Complex of Costa Rica [*Schmidt-Effing*, 1980] appear virtually identical to faunas (MH-7) within the Marin Headlands terrane of northern California [*Murchey*, 1984]. Although our model of Franciscan-type chert deposition within the equatorial zone of high productivity is supported by the available paleomagnetic evidence, more data are needed from both deep-marine and near-shore tuffaceous cherts to determine the Mesozoic paleogeography of chert deposition within the Pacific Basin.

Blocks of Lower Jurassic radiolarian chert are also exposed in the San Rafael Mountains of southern California (Figure 1). Most of the chert outcrops there are recrystallized, but we sampled moderately fine-grained relatively unaltered rocks to further test our model at two localities: one near Goat Rock and the second in Redrock Canyon. Samples were also collected from overlying Upper Cretaceous sedimentary rocks of the Cachuma Formation in Agua Caliente Canyon to determine an initial paleolatitude for these continentally derived rocks, thereby constraining the paleolatitude of accretion for the underlying chert blocks. These new data are compared with paleomagnetic results for similar Upper Cretaceous sedimentary rocks in the region at Figueroa Mountain [*McWilliams and Howell*, 1982] and with Middle to Upper Jurassic pillow basalts and red tuffaceous cherts of the Coast Range ophiolite at Stanley Mountain [*McWilliams and Howell*, 1982; *Hagstrum and Murchey*, 1995]. In addition, an updated summary of the

[1]Now at Century West Engineering Corporation, Dublin, California.

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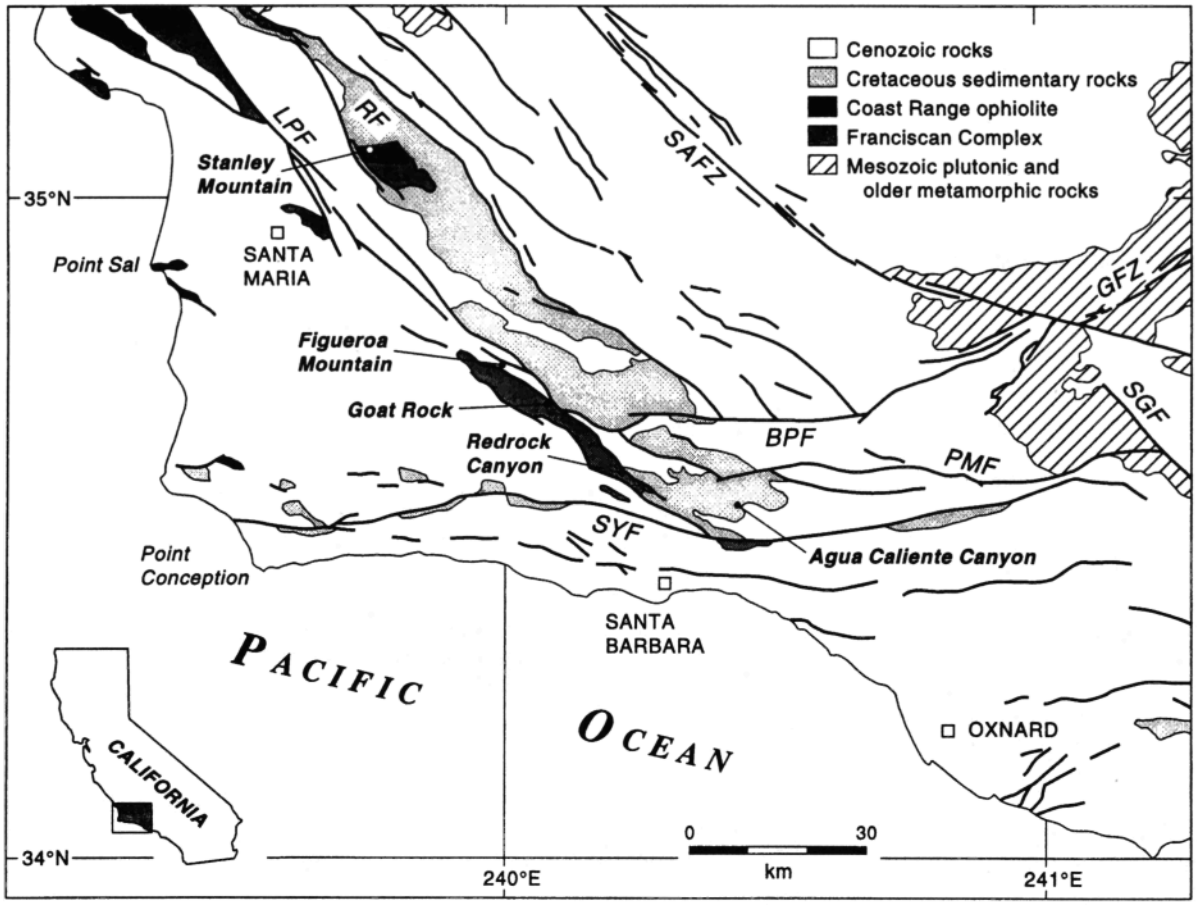


Figure 1. Map of southern California (see inset) showing paleomagnetic sampling locations of Lower Jurassic radiolarian chert within the Franciscan Complex (Goat rock and Redrock Canyon, this study), Upper Cretaceous sedimentary rocks (Agua Caliente Canyon, this study; Figueroa Mountain, *McWilliams and Howell* [1982]), and Upper Jurassic Coast Range ophiolite at Stanley Mountain [*McWilliams and Howell*, 1982; *Hagstrum and Murchey*, 1995]. Fine lines indicate geologic contacts, and heavy lines represent faults (mostly right-lateral strike-slip): SAFZ, San Andreas fault zone; GFZ, Garlock fault zone; LPF, Little Pine fault; RF, Rinconada fault; BPF, Big Pine fault; PMF, Pine Mountain fault; SGF, San Gabriel fault; SYF, Santa Ynez fault.

paleomagnetic data for Franciscan-type cherts in northern California and Mexico is presented.

Geologic Setting

Rocks of the Franciscan Complex crop out in the San Rafael Mountains, near their southern terminus in California (Figure 1), as a mélange belt (1.5–4 km wide) that contains blocks of graywacke, greenstone, chert, serpentinite, and other minor rock types in a matrix of strongly sheared mudstone and serpentinite [*Wahl*, 1995]. Located in the allochthonous Nacimiento block, these Franciscan rocks are separated from coeval forearc basin rocks of the Espada Formation by a fault contact [*Dibblee*, 1966]. The Camuesa and Little Pine faults form northern and southern boundaries of the mélange belt and separate it from strongly folded Cretaceous, Paleogene, and Neogene sedimentary rocks. These younger formations have fold axes parallel to the trend of the mélange belt suggesting that all rocks in this region were subjected relatively recently to north-northeast shortening [*Warner*, 1992].

The Franciscan Complex in California is a remnant of the accretionary wedge formed by subduction along the western margin of North America during Jurassic through early Cenozoic time [*Blake and Jones*, 1978; *Page*, 1981]. Based on a westward decrease in age and metamorphic grade, the Franciscan Complex has been divided into three fault-bounded belts: (1) the Eastern belt, present only in the northern Coast Ranges, consisting of Middle Jurassic to Upper Cretaceous rocks metamorphosed to the blueschist facies; (2) the Central belt, extending from northernmost California southward through the southern Coast Ranges, comprising a Lower Jurassic to Paleocene tectonic mélange with pumpellyite as the dominant metamorphic mineral within the graywacke matrix; and (3) the Coastal belt, found (onshore at least) only north of San Francisco, containing Upper Cretaceous to Miocene low-grade rocks characterized by laumontite veins [*Blake et al.*, 1988; *McLaughlin et al.*, 1988]. Franciscan rocks of the Nacimiento block most closely resemble those of the Central belt [*Page*, 1981].

Early Jurassic radiolarians were first recognized in bedded chert of the Franciscan Complex by *Pessagno and Blome* [1980] and *Pessagno and Poisson* [1981]. Pessagno's 11-m section, called the "San Rafael" section, is our "Goat Rock" locality, and was initially considered to be Late Jurassic in age [*Pessagno*, 1977]. The age assignment was revised when the Goat

Rock faunas were compared to newly discovered faunas from molluscan-bearing Lower Jurassic rocks of Turkey and North America [*Pessagno and Blome*, 1980; *Pessagno and Poisson*, 1981; *Pessagno and Whalen*, 1982], including the late Pliensbachian faunas from the Nicely Formation of eastern Oregon.

Radiolarians observed within the Goat Rock section have been listed by previous workers separately in several different publications; they are presented together here for the first time. Taxa occurring in the interval between 4 and 10 m above the basal chert-basalt contact include *Pantanellium sanrafaelense* Pessagno and Blome, *Canoptum anulatum* Pessagno and Poisson, and *Canoptum rugosum* Pessagno and Poisson. A sample collected by Pessagno from the top of his 11-m section contains those radiolarians as well as *Pobum infinitum* (Pessagno and Poisson), *Wrangellium* sp., *Canutus izeensis* Pessagno and Whalen, *Praeconocaryomma media* Pessagno and Poisson, *Orbiculiforma mclaughlani* Pessagno, *Bagotum modestum* Pessagno and Whalen, *Droltus* sp., *Praeconocaryomma parvimamma* Pessagno and Poisson, *Protopsium* sp., and *Pseudocrucella sanfilippoae* (Pessagno) [*Pessagno and Blome*, 1980; *Pessagno and Poisson*, 1981; *Pessagno and Whalen*, 1982].

We sampled the chert interval from 7.0 m to 18.65 m above the basal contact of chert and basalt at Goat Rock for paleomagnetic and biostratigraphic analysis. The previously unstudied upper part of the section also contains late Pliensbachian radiolarian faunas including *Canoptum anulatum*, *Praeconocaryomma* sp. aff. *P. whiteavesi* Carter, *Paronaella* (*Sontonaella*) spp., *Canutus* spp., and less commonly *Katroma ninstintsi* Carter, *Droltus* sp., and *Hagiastrum* sp. cf. *H. egregium* Rüst. The sampled interval at Goat Rock (from ~4 m to ~19 m) best correlates with the middle part of Franciscan Marin Headlands assemblage MH-1 of *Murchey* [1984] in the Marin Headlands and at Bald Mountain near Mount Umunhum [*Hagstrum and Murchey*, 1993]. The Goat Rock section is late Pliensbachian in age.

The radiolarian faunas from the Redrock section have not been previously reported. From the basal contact with underlying basalt, we collected microfossil samples from about 20 m of section. *Canoptum* spp., *Lupherium* spp., and *Praeconocaryomma* spp. occur in samples throughout the section. *Pantanellium sanrafaelense* occurs through the lower 17 m of section but is replaced by *Trillus elkhornensis* Pessagno and Blome at the top of the section (19.9 m). *Canutus* spp., *Katroma* spp. and *Paronaella* (*Sontonaella*) spp. occur in the interval above 9 m. The Redrock section is correlative with the lower?, middle, and upper part of assemblage MH-1 of *Murchey* [1984] in the Marin Headlands and at Bald Mountain near Mount Umunhum [*Hagstrum and Murchey*, 1993]. The Redrock section ranges in age from late(?) Pliensbachian to late Pliensbachian or early to middle Toarcian.

The Espada Formation exposed in Agua Caliente Canyon contains fossils of latest Jurassic and Early Cretaceous age and is composed primarily of dark greenish-gray mudstone and siltstone with minor sandstone and rare conglomerate [*Dibblee*, 1966, 1986]. These rocks are stratigraphically equivalent to the Great Valley sequence east of the San Andreas fault [*Dickinson and Rich*, 1972], formed from turbidites shed off the inboard magmatic arc, and represent a submarine fan depositional environment [*MacKinnon*, 1976]. Disconformably overlying the Espada Formation in many areas of the San Rafael Mountains are the shale and sandstone members of the Upper Cretaceous (primarily Campanian) Cachuma Formation [*MacKinnon*, 1976; *Dibblee*, 1991]. The shale member is grayer in color and generally lacks carbonaceous flakes, and the sandstone member is tan to light gray in color and has a higher percentage of sandstone, than similar rock in the Espada Formation [*Dibblee*, 1966]. The contact between the Espada Formation and the Cachma Formation is disconformable, and these rocks in turn are disconformably overlain by the Eocene Juncal Formation in Agua Caliente Canyon [*MacKinnon*, 1976].

Paleomagnetic Methods

Sequences of Lower Jurassic radiolarian chert were collected for paleomagnetic analysis at cliff-face exposures ~1.5 km east-northeast of Goat Rock (34.7°N, 240.1°E) on the Figueroa Mountain quadrangle (1:24,000 scale) and in Redrock Canyon ~2 km west-southwest of Alexander Peak (34.6°N, 240.3°E) on the San Marcos Pass quadrangle (Figure 1). Samples were collected as oriented blocks at 1- to 5-m intervals (Figure 2) from brownish-red to purplish-red layers thick enough (>5 cm) for laboratory drilling. Zones of alteration (green chert) generally found toward the tops of the chert sections, and zones of recrystallization and manganese staining found toward their bases, were avoided. Bedding attitudes were measured at each sample level, and due to the similarity of attitudes at each locality, an average of these measurements (Table 1) was used for the overall structural corrections (rotation of the bedding planes to horizontal about the strike axis). The basalt-chert contact is preserved at the bases of both chert sequences, but the upper parts are truncated by faulting. The two sampled sequences of late Pliensbachian to middle Toarcian age are between 10 and 20 m thick, indicating a postcompaction rate of chert deposition of ~2.5 m/m.y.

A ~30-m-thick section of Upper Cretaceous (Campanian) fine sandstones and siltstones of the Cachuma Formation (Figure 2) was also collected in Agua Caliente Canyon. The uppermost 4 m is separated from the lower 20 m of the sampled section by a 10-m covered interval. The sampling site is ~100 m north-northwest of the Agua Caliente spring (34.5°N, 240.4°E) on the Hildreth Peak quadrangle [see *Dibblee*, 1986]. Between two and five oriented drill cores were taken per bedding layer. Again attitudes of each sampled layer were measured, and due to their similarity these measurements were averaged to make an overall structural correction.

Chert specimens were subjected to thermal demagnetization up to ~680°C (Figures 3 and 4), and acquisition of isothermal remanent magnetization (IRM) experiments was conducted in fields up to 700 mT on selected specimens to help identify their magnetic mineralogy (Figure 5). Both alternating field (af) demagnetization, using peak inductions of up to 100 mT, and thermal demagnetization to ~500°C were performed on the Agua Caliente specimens (Figure 6). Remanent magnetizations of all specimens were measured using a cryogenic magnetometer. A saturation magnetization (J_s) versus temperature (T) experiment was also conducted on a magnetic separate from 12 specimens of the Cachuma Formation to help identify the magnetic mineralogy of these clastic rocks (Figure 6).

Components of magnetization isolated by thermal and af demagnetization were fitted with least squares lines [*Kirschvink*, 1980], and the site-mean and overall-mean directions (Tables 1 and 2) are plotted in Figures 7 and 8. Individual components are defined by line segments of three or more

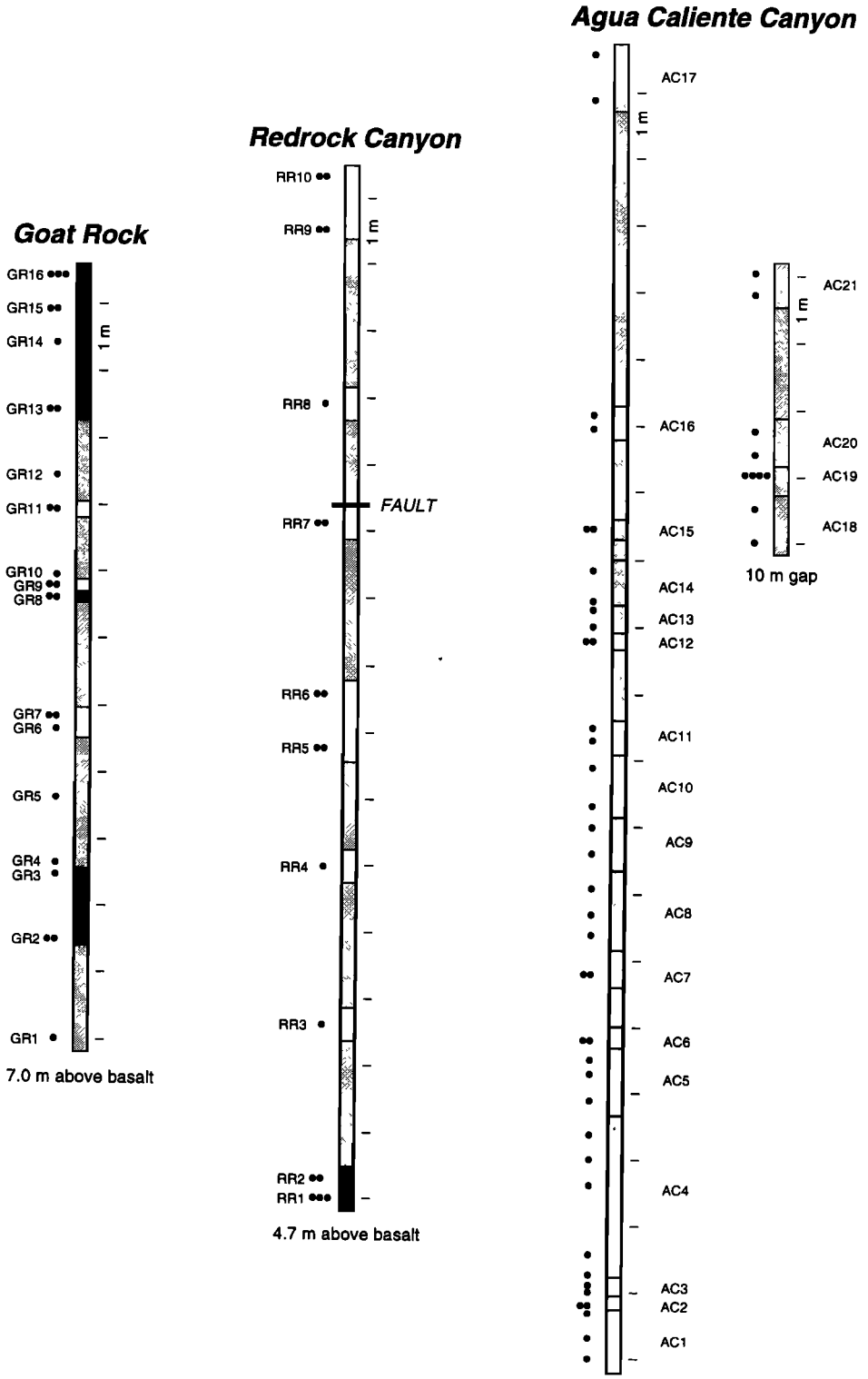


Figure 2. Sections of Lower Jurassic radiolarian chert near Goat Rock and in Redrock Canyon, and Upper Cretaceous siltstones and fine sandstones of the Cachuma Formation in Agua Caliente Canyon, showing relative sampling positions (dots). For the chert sections. labels indicate the number of hand samples collected, and the number of dots indicates the number of specimens drilled from each sample. Black zones indicate normal(?)-polarity *C* component magnetizations, white zones reversed(?) polarity, and gray zones are indeterminate. Sample GR1 is 7.0 m above the basalt-chert contact, and sample RR1 is 4.7 m above the basalt-chert contact. For the Agua Caliente Canyon section, labels indicate the number of beds sampled, and multiple dots per horizon indicate the number of specimens per sample analyzed. Scale interval is 1 m. There is a 10-m gap (covered section) between beds AC17 and AC18.

Table 1. Paleomagnetic Data for Radiolarian Chert Within the Franciscan Complex, San Rafael Mountains, California

Site	I_t, deg	D_t, deg	I_c, deg	D_c, deg	N	R	k	α_{95}, deg	λ_{PAL}, deg	λ_{VGP}, deg	ϕ_{VGP}, deg
					Component C						
Goat Rock											
GR16	23.0	301.3	10.8	312.6	3	2.9787			5	32.4	142.1
GR15	24.8	301.2	12.3	313.8	2	1.9956			6	32.9	143.1
GR14	27.8	305.1	12.5	318.5	1				6	42.4	123.3
GR13	33.7	284.1	28.8	308.0	2	1.9993			15	21.7	158.0
GR11	−20.7	128.6	−4.9	136.8	2	1.9946			2	37.6	135.8
GR9	−18.6	131.4	−1.7	137.8	2	1.9988			1	39.1	132.5
GR8	37.6	291.5	27.4	316.0	2	1.9949			15	29.1	156.7
GR7	−19.8	123.2	−7.2	132.2	2	1.9968			4	33.0	139.0
GR6	−26.1	131.7	−7.7	142.3	1				4	43.4	117.3
GR3	27.1	290.8	20.0	307.8	1				10	36.7	135.9
GR2	24.6	298.5	13.6	311.7	2	1.9996			7	30.6	144.7
Avg N			14.8	323.2	(9)	8.3691	13	15.0		46.7	120.0
Avg R			5.4	317.3	(4)	3.9888	269	5.6		39.1	120.8
Avg all	26.0	301.0	13.3	314.4	(11)	10.8560	69	5.5	7 ± 3	39.7	127.3
Redrock Canyon											
RR10	−33.8	165.1	12.6	179.6	2	1.9984			6	69.2	103.7
RR9	−29.0	165.1	14.6	175.6	2	1.9966			7	66.7	99.0
RR8	−26.8	149.7	2.6	168.3	1				1	55.9	121.0
RR7	−31.6	154.1	4.8	174.0	2	1.9991			2	61.0	119.7
RR6	−32.5	170.1	16.9	180.3	2	1.9945			9	70.9	90.4
RR5	−27.0	152.1	4.6	169.1	2	1.9998			2	57.6	118.4
RR4	−36.0	179.3	21.5	187.7	1				11	75.4	62.9
RR3	−22.1	177.2	27.5	173.7	1				15	66.7	67.2
RR2	34.0	324.3	3.7	354.1	2	1.9782			2	54.6	132.8
RR1	39.9	337.7	−4.8	2.8	3	2.9681			2	67.1	124.9
Avg N			−0.6	358.5	(2)	1.9888				55.1	63.0
Avg R			−13.2	355.9	(8)	7.8767	57	7.4		48.5	66.4
Avg all	31.8	341.5	−10.7	356.4	(10)	9.8247	51	6.8	5 ± 4	49.9	65.9
Basalt[a]	70.1	318.9	12.6	29.6	21	19.7150	16	8.3	6 ± 3[b]	50.7	9.4
Avg-all (GR + RR)			2.1	334.4	(21)	18.8871	9	10.9	1 ± 4[b]	48.8	101.1
					Component B						
Goat Rock											
GR16	71.0	205.9	70.7	12.2	3	2.8613			55	67.9	258.8
GR15	73.0	217.5	67.6	4.9	2	1.9997			50	73.8	251.3
GR12	50.5	194.0	86.5	132.6	2	1.9999			83	29.8	246.0
GR10	51.7	175.2	75.4	100.7	1				62	25.7	270.4
GR8	67.0	211.4	73.8	1.5	2	1.9901			60	64.8	241.9
GR4	67.9	180.1	71.7	41.8	2	1.9985			57	54.4	279.3
GR3	61.0	172.0	72.8	67.0	2	1.9880			58	40.8	279.9
GR1	61.8	144.2	59.6	68.7	1				40	36.6	302.2
Avg	64.7	184.9	75.4	43.6	(8)	7.8138	38	9.2	62 ± 16	51.3	270.6
Redrock Canyon											
RR10	30.2	178.9	39.9	111.3	2	1.9973			23	3.3	119.7
RR9	29.9	195.4	53.8	106.6	2	1.9984			34	7.3	293.2
RR8	46.2	239.1	55.9	38.7	3	2.9992			36	58.7	315.5
RR7	26.3	202.9	61.3	108.3	2	1.9996			42	11.1	285.9
RR6	50.3	202.2	46.4	75.6	2	1.9986			28	26.4	313.6
RR4	48.9	217.2	52.6	63.9	2	1.9864			338	37.9	312.5
Avg	40.1	204.2	54.7	85.2	(6)	5.7272	18	16.1	35 ± 16	22.6	302.2
					Component B'						
Goat Rock											
GR11	26.1	51.0			4	3.9399	50	13.1	14 ± 8	39.7	341.5
GR10	33.8	73.5			2	1.9944			19	23.7	323.4
GR9	44.5	28.5			2	1.9995			26	64.1	341.6
GR7	48.4	50.4			3	2.9882			29	47.4	322.7
GR6	21.6	51.7			1				11	37.6	343.7
Avg	35.7	51.7			(5)	4.8238	23	16.4	20 ± 12	42.2	334.3
					Component A						
Goat Rock											
GR17	75.5	3.5			1				63	62.0	243.5
GR16	72.9	2.8			3	2.9932			58	66.2	243.7
GR15	66.3	348.5			2	1.9995			49	73.6	212.3
GR14	64.4	16.3			1				46	73.1	282.1
GR13	62.2	11.6			2	1.9972			43	77.4	282.1
GR12	48.2	16.5			2	1.9897			29	75.0	347.1
GR11	64.9	13.0			2	1.9633			47	74.4	274.9
GR10	62.2	325.0			1				43	61.7	178.8

Table 1. (continued)

Site	I_t, deg	D_t, deg	I_c, deg	D_c, deg	N	R	k	α_{95}, deg	λ_{PAL}, deg	λ_{VGP}, deg	ϕ_{VGP}, deg
					Component A (continued)						
Goat Rock (continued)											
GR8	66.6	355.0			2	1.9806			49	75.1	227.4
GR7	69.3	23.0			3	2.9860			53	65.6	274.8
GR6	59.6	18.5			1				40	74.3	303.2
GR4	64.3	20.3			2	1.9862			46	70.9	287.4
Avg	65.6	7.0			(12)	11.8349	67	5.4	48 ± 7	75.9	259.8
Redrock Canyon											
RR9	67.5	7.5			2	1.9883			50	73.3	257.2
RR7	75.0	341.3			1				62	60.3	222.5
RR6	64.5	8.0			2	1.9999			46	76.7	265.0
RR5	48.6	19.5			1				30	72.8	341.9
RR4	48.9	4.3			1				30	84.0	21.9
RR2	61.5	320.5			1				43	58.4	177.0
Avg	62.5	359.9			(6)	5.8360	30	12.3	44 ± 15	80.8	239.7

Site, near Goat Rock (GR; 34.7°N, 240.1°E); in Redrock Canyon (RR; 34.6°N, 240.3°E); I_t and D_t are in situ inclination and declination of paleomagnetic directions (I_c, D_c, corrected for structural tilt; strike/dip: GR, 289°/38°; RR, 320°/77°; dip is 90° clockwise of strike); N is number of specimens (number of bedding layers given in parentheses); R is vector sum of N unit vectors; k is the concentration parameter [*Fisher*, 1953]; α_{95} is the radius of 95% confidence; λ_{PAL} is paleolatitude with averaged 95% confidence limits; λ_{VGP} and ϕ_{VGP} are north latitude and east longitude, respectively, of virtual geomagnetic pole (components B and C, corrected for tilt; components A and B', in situ). Avg, average.

[a]Paleomagnetic data for pillow basalts underlying the radiolarian chert at Redrock Canyon provided by D. Smith et al. (personal communication, 1993). Not included in the overall mean; see text.

[b]Paleolatitude and 95% confidence based on inclinations only after *McFadden and Reid* [1982].

points on orthogonal vector plots (Figures 3, 4, and 6). Although the observed components might be composite due to overlapping unblocking temperatures of two or more unresolved components, line segments are generally interpreted as single components of magnetization. The chert blocks sampled for this investigation are the least altered of those we have studied for paleomagnetism, and the highest-temperature components appear exceptionally well preserved.

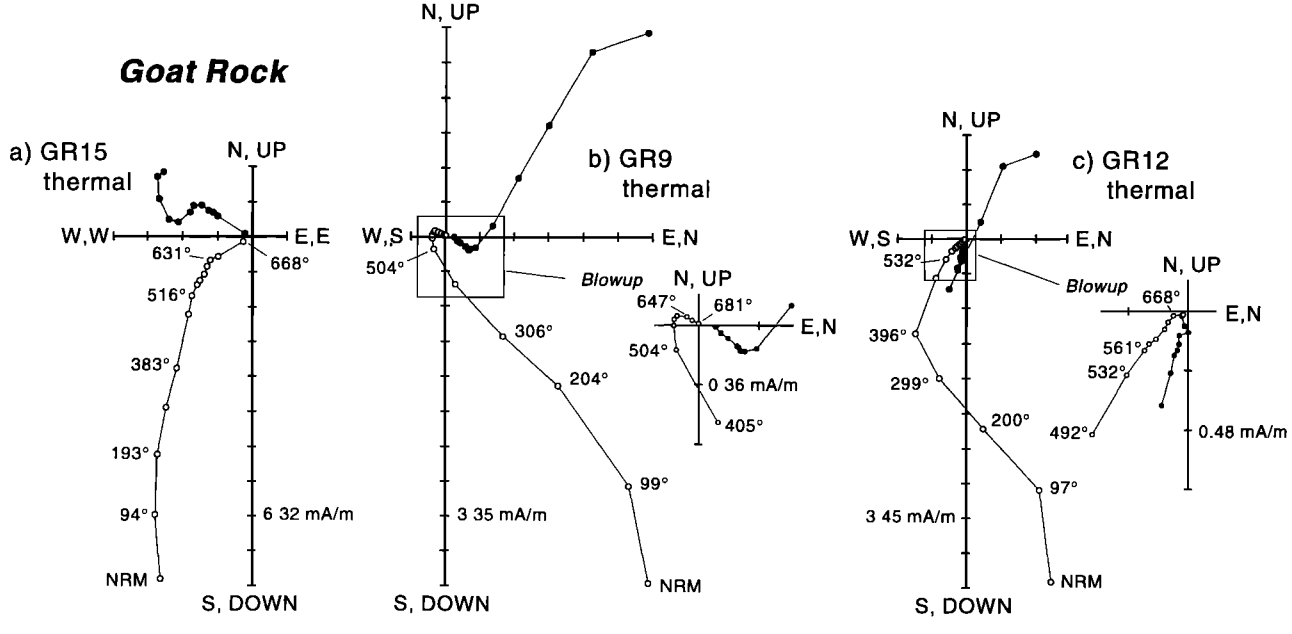


Figure 3. Orthogonal projections (in situ) of thermal demagnetization vector endpoints for chert specimens from the section near Goat Rock: solid circles, horizontal plane; open circles, vertical plane. (a) A specimen from sample GR15 shows sequential removal of components A (NRM to 193°C), B (383° to 516°C), and normal(?)-polarity C (631° to 668°C). (b) A specimen from sample GR9 shows removal of components B' (99° to 504°C) and reversed(?)-polarity C (647° to 681°C). (c) The specimen from sample GR12 shows removal of components B' (97° to 299°C), B (396° to 532°C), and apparently reversed(?)-polarity C (561° to 668°C) at the uppermost demagnetization steps (see blowup).

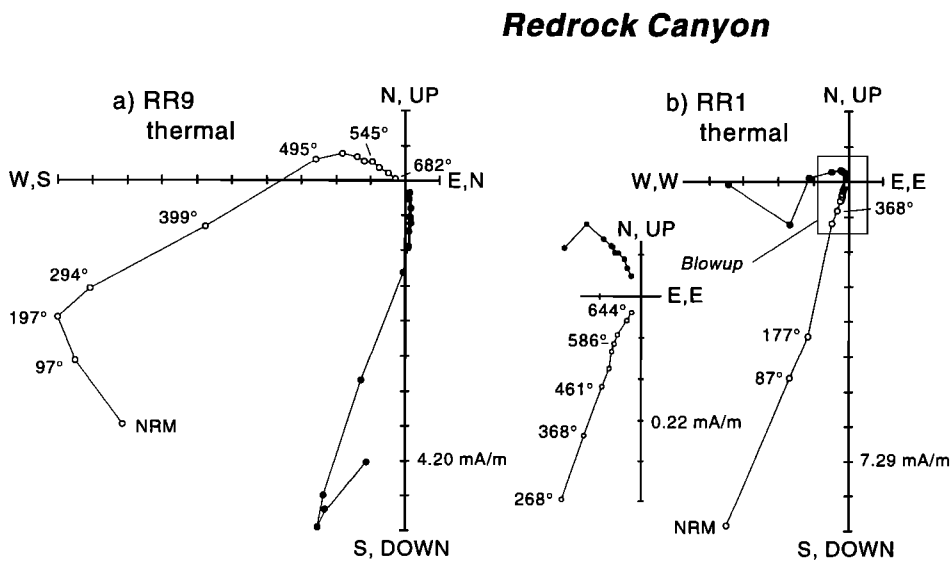


Figure 4. Orthogonal projections (in situ) of thermal demagnetization vector endpoints for Redrock Canyon chert specimens: solid circles, horizontal plane; open circles, vertical plane. (a) A specimen from sample RR9 shows removal of components *A* (NRM to 197°C), *B* (197° to 495°C), and reversed(?)-polarity *C* (623° to 682°C). (b) A specimen from sample RR1 show removal of components *B* (87° to 368°C) and normal(?)-polarity *C* (586° to 644°C).

Discussion

Franciscan-Type Cherts

Four separate components of remanent magnetization were isolated in the chert samples during thermal demagnetization. The first component (*A*) removed (Figure 3a), usually by 300°C, has an in situ direction similar to that of the present-day field (I = 59.3°, D = 14.5°) and recent axial-dipole field (I = 54.1°, D = 0.0°) for southern California (Figure 7). This component could be either a recently acquired thermoviscous magnetization, one related to deep burial (several kilometers) followed by regional uplift in the late Tertiary [see *Pullaiah et al.*, 1975], or a composite of both.

The second component (*B'*) removed (Figures 3b and 3c), usually between 100° and 500°C, is found only in samples from Goat Rock. The *B'* component (Figure 8) is similar to the overprint direction observed in pillow basalts and overlying Upper Jurassic sedimentary rocks of the Coast Range ophiolite at Stanley Mountain [*McWilliams and Howell*, 1982; *Hagstrum*, 1992; *Hagstrum and Murchey*, 1995]. This component is interpreted as an overprint magnetization and its direction possibly indicates clockwise vertical-axis rotation (~50°) of the Stanley Mountain and Goat Rock block similar to those documented in the western Transverse Ranges [*Hornafius et al.*, 1986], assuming that these rocks have not been significantly tilted since the Miocene. The *B'* magnetization, therefore, might have been acquired during early Miocene intrusive and volcanic activity prior to clockwise rotation in the region.

The third component (*B*) isolated, mostly between 200° and 600°C (Figures 3a, 3c, and 4), is the dominant component found in most of the chert samples. The age of the *B* magnetization is also uncertain, and if young (post-tilting), it could imply clockwise rotation of the chert blocks ~195°. If acquired prior to structural tilting, possibly as a chemical overprint during subduction and accretion of the chert to the continental margin, the *B* component could represent a clockwise rotated (~45° to 85°) Cretaceous(?) magnetization (Table 1). The attitude at which the chert blocks were overprinted, however, is indeterminate, so no clear interpretation of these secondary components can be made. If component *B* is related to accretion, it does not indicate a low paleolatitude after structural correction as has been proposed for other accretion-related secondary magnetizations in similar rocks [*Hagstrum*, 1990, 1992].

The fourth and last component (*C*) removed from the chert samples, generally above ~620°C, has dual-polarity directions (Figure 2 and Table 1) which are interpreted to represent primary magnetizations acquired as, or soon after, the chert was deposited in both normal- and reversed-polarity geomagnetic fields. Data for both sections pass reversal tests at the 'C' level [*McFadden and McElhinny*, 1990]. The possibility that this polarity stratigraphy is related to variable overprinting in these rocks is ruled out because the secondary components (*A*, *B'*, *B*) have different directions than the *C* component directions. Samples from the chert sequence in Redrock Canyon mostly have southerly declination magnetizations (Figure 2) which may be due to random sampling where zones with northerly declination magnetizations were missed, possibly combined with fault repetition of section.

The Redrock chert samples also have a significantly different characteristic magnetization than the underlying pillow basalts (D. Smith et al., personal communication, 1993). This could be due to differential vertical-axis rotation on a bedding plane fault or to incomplete separation of the *B* and *C* components during demagnetization in the pillow basalt samples. At three other localities of Franciscan-type chert in northern California [*Hagstrum and Murchey*, 1993] and Mexico [*Hagstrum and Sedlock*, 1990] where the basalt-chert contact is exposed, the underlying pillow basalts have been completely overprinted by the *B* magnetization. Because of this uncertainty, results for the pillow basalts in Redrock Canyon are not included in the overall mean (Table 1).

Both the thermal demagnetization and IRM experiments indicate that hematite is the predominant magnetic mineral in

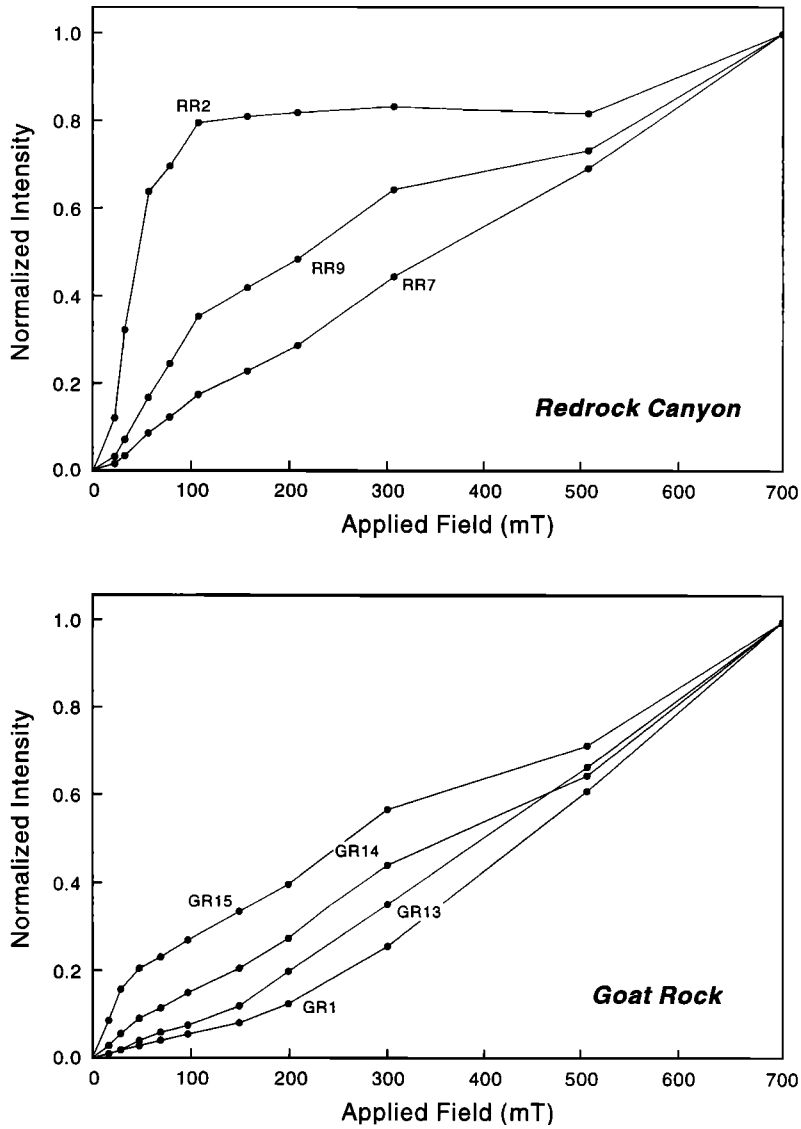


Figure 5. Normalized isothermal remanent magnetization (IRM) plots for chert specimens from (top) Redrock Canyon and (bottom) near Goat rock. Rapid saturation of RR2 indicates the presence of a cubic phase which might be related to magnesium oxide staining toward the base of this section resulting from seafloor hydrothermal alteration. Other specimens from Redrock Canyon and near Goat rock indicate that hematite is the dominant remanence carrier in these rocks. Specimens RR9, GR14, and GR15 show the presence of slight amounts of a cubic phase (magnetite?) which is probably related to upsection alteration of the red chert to green chert.

the chert specimens and that it is the carrier of the *C* component of magnetization. Some specimens (GR15, RR9; Figure 5) show a more rapid increase in magnetization below 100 mT, indicating higher quantities of a cubic phase like magnetite or maghemite. These samples are near the tops of the sampled sections where the cherts are more altered to green chert. The rapid increase of magnetization in specimen RR2, collected near the base of the Redrock section, might instead be due to the presence of manganese oxides deposited during seafloor alteration.

After structural correction, component *C* has shallow inclinations (Figure 8 and Table 1) indicating acquisition at equatorial paleolatitudes. Although a fold test cannot be applied to the two San Rafael chert blocks, the agreement of their corrected *C* inclinations (paleolatitudes) with the coeval chert blocks sampled in northern California and Mexico (Figure 9b) indicates that the *C* component is an early pretilt magnetization. This consistency also suggests that these San Rafael cherts were formed together with the northern California cherts on the Farallon plate beneath the equatorial zone of high biologic productivity (Figure 10). The San Rafael cherts were also probably accreted at low paleolatitudes (~15°N) and were likely strung out along the margin by subsequent strike-slip faulting as well [*Hagstrum and Murchey*, 1993]. The paleolatitude of accretion, however, is determined from only six samples of latest Jurassic and Early Cretaceous chert in the Marin Headlands terrane and is not well constrained (15° ± 12°N). Accretion from Central America to southern Mexico also satisfies models of Farallon plate motion starting from equatorial paleolatitudes in the Early Jurassic (Figure 9).

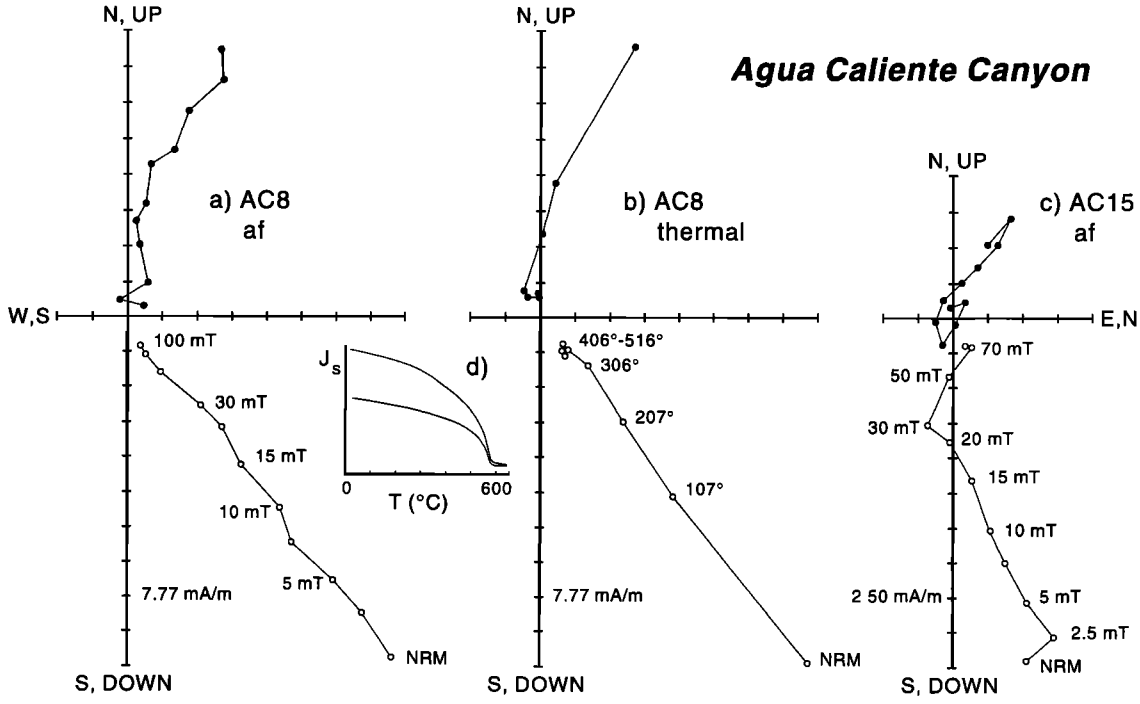


Figure 6. Orthogonal projections (in situ) of alternating field (af) and thermal demagnetization vector endpoints for fine-sandstone and siltstone specimens of the Cachuma Formation in Agua Caliente Canyon: solid circles, horizontal plane; open circles, vertical plane. (a) and (b) The af and thermal demagnetization of component *A* (NRM to 100 mT; NRM to 406°C). (c) Removal of components *A* (2.5–20 mT) and *B* (30–70 mT); component *B* is rarely isolated in these rocks. (d) Saturation magnetization (J_s) is plotted against temperature for a magnetic separate made from 12 specimens from beds AC1, AC2, AC4, AC5, AC8, and AC15. A Curie temperature of ~580°C indicates that Ti-poor magnetite is the remanence carrier in these rocks.

Deposition of Franciscan-type chert sequences within the equatorial zone of high biologic productivity (Figure 10) is supported by rough estimates of their sedimentation rates. Siliceous sediments along the equator currently accumulate at 4–5 cm per 1000 years (40–50 m/m.y.) [*Kennett*, 1982]. Assuming that dewatering and compaction associated with burial and accretion would reduce unconsolidated high-porosity sediments to about 10% of original thickness (or 4–5 m/m.y.),

Table 2. Paleomagnetic Data for Upper Cretaceous Sedimentary Rocks of the Cachuma Formation in Agua Caliente Canyon, Santa Rafael Mountains, California, for Component A

Site	I_t, deg	D_t, deg	I_c, deg	D_c, deg	N	R	k	α_{95}, deg	λ_{PAL}, deg	λ_{VGP}, deg	ϕ_{VGP}, deg
AC21	61.7	24.1	17.7	39.7	2	1.9959			43	69.5	299.2
AC20	54.3	28.9	20.7	41.1	2	1.9978			352	66.3	321.3
AC19	53.3	10.8	18.7	41.9	4	3.9934	456	4.3	34 ± 4	81.1	331.6
AC18	55.7	350.7	21.5	35.3	2	1.9963			36	82.2	166.1
AC17	59.2	12.9	23.7	44.2	2	1.9387			40	78.4	298.6
AC16	60.0	35.8	21.5	51.9	2	1.9916			41	61.1	306.8
AC15	61.1	19.6	17.5	44.0	2	1.9952			42	72.9	298.0
AC14	60.8	13.7	21.5	50.0	2	1.9913			42	77.0	292.3
AC13	57.5	22.6	14.3	42.2	2	1.9802			38	71.5	312.6
AC12	66.5	13.3	23.2	33.2	2	1.9939			49	72.5	270.7
AC11	53.3	357.4	25.5	49.8	2	1.9975			34	87.7	134.4
AC10	53.4	18.1	15.9	46.9	2	1.9784			34	75.1	327.3
AC9	63.7	20.1	21.4	45.5	2	1.9758			45	71.3	289.1
AC8	57.0	16.8	19.7	48.0	3	2.9933			38	76.1	312.9
AC7	64.8	23.3	14.3	54.3	2	1.9864			47	68.7	288.6
AC6	61.5	8.6	20.5	44.0	2	1.9662			43	79.5	277.3
AC5	53.8	1.7	27.8	33.0	3	2.9499			34	88.6	335.8
AC4	56.5	12.7	17.2	38.8	5	4.9832	238	5.0	37 ± 5	79.4	312.5
AC3	57.4	9.0	11.3	48.0	2	1.9990			38	81.9	302.2
AC2	56.9	6.3	21.5	39.7	2	1.9955			37	84.1	298.3
AC1	52.5	14.3	15.1	39.8	3	2.9900			33	78.1	333.1
Avg	58.6	13.9	19.7	44.0	(21)	20.8564	139	2.7	39 ± 3	77.9	303.2

Parameters as in Table 1. Site location is 34.5°N, 240.4°E; strike/dip are 342°/50°; paleolatitudes (λ_{PAL}) and VGPs are in situ.

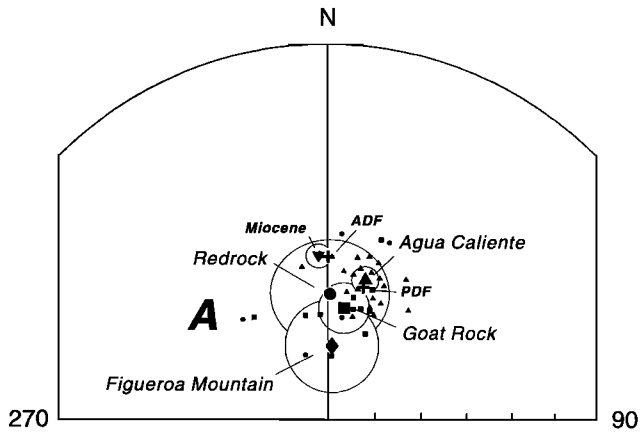


Figure 7. Equal-area plot of mean in situ directions for the *A* component from Lower Jurassic chert samples near Goat Rock (square) and in Redrock Canyon (circle) and from Upper Cretaceous sandstone and siltstone of the Cachuma Formation in Agua Caliente Canyon (inverted triangle) and at Figueroa Mountain (diamond) [*McWilliams and Howell*, 1982]. Smaller symbols indicate sample/bedding layer means. Triangle indicates expected Miocene direction in southern California (I = 53.2°, D = 357.1°) using the reference pole of *Hagstrum et al.* [1987], and crosses indicate the present-day field (PDF; I = 59.3°, D = 14.5°) and axial-dipole field (ADF; I = 54.1°, D = 0.0°) directions. Solid symbols indicate lower hemisphere points.

these accumulation rates are comparable to those calculated for Early and Middle Jurassic chert of the Franciscan Complex (averaging ≥2 m/m.y. post compaction). Brown clays presently accumulate beneath low-fertility control water masses at rates of <2 mm per 1000 years (<2 m/m.y. uncompacted) [*Kennett*, 1982]. Again, assuming compaction and dewatering to 10% of the original thickness, these accumulation rates might translate to 0.2 m/m.y. in accreted strata, comparable to postcompaction accumulation rates of about 0.14 m/m.y. calculated for MH-5 to MH-7 in the Marin Headlands and The Geysers sequences.

New biostratigraphic analyses of the paleomagnetic specimens having *C* component magnetizations from Cedros Island and the San Benito Islands in Baja California, Mexico [*Hagstrum and Sedlock*, 1992], indicate that they have distinctly different ages. The Cedros specimens correlate with the MH-2 biozone and have an average paleolatitude of 3° ± 4° north or south (I = 5.3°, k = 120, N = 10), and the San Benito specimens correlate with the MH-6 biozone and have an average paleolatitude of 2° ± 4° north or south (I = 3.7°, k = 246, N = 6) based on the inclination-only statistics of *McFadden and Reid* [1982]. These results have been plotted on Figure 9 and clearly indicate different plate motion trajectories for the two chert sequences because of their deposition within the equatorial productivity zone at different times, possibly on different plates.

This separation in time and space between the Baja California cherts is also corroborated by lithology of the two sequences [*Sedlock and Isozaki*, 1990]. The Cedros cherts are represented by a thick (~40 m) homogenous sequence of Late Triassic to Early Cretaceous (≤80 m.y.) ribbon chert which contains very minor amounts of terrigenous or volcanogenic sediments. These rocks were deposited in a open ocean environment similar to the Franciscan chert sequences in northern and southern California. The San Benito sequence, on the other hand, contains ribbon chert and significant amounts of volcanogenic sediments (pebbly sandstones) and volcanic megabreccia (limestone, chert, metabasite blocks). These rocks were interpreted by *Sedlock and Isozaki* [1990] to indicate deposition at the foot of a volcanic edifice on ocean floor undergoing extension and normal faulting as it entered a subduction zone. Parenthetically, the red tuffaceous chert of the Coast Range ophiolite at Stanley Mountain was also deposited in a different environment than the Franciscan cherts (Figure 9), most likely in a productivity high along the western margin of North America near 32° ± 8°N paleolatitude [*Hagstrum and Murchey*, 1995] (Figure 10).

Upper Cretaceous Section

MacKinnon [1976] collected 40 samples of the Campanian Cachuma Formation in Agua Caliente Canyon for paleomagnetic analysis. Anisotropy of magnetic susceptibility (AMS) was also measured in twenty of these samples. Current structures (63) measured in Agua Caliente Canyon indicated a mean current direction to the south-southeast (168°; s.d. = 22°), similar to the maximum AMS direction. Paleomagnetic results are widely scattered after af demagnetization showing corrected characteristic directions with northeast declinations and shallow to moderate inclinations.

The af demagnetization of specimens collected from the Cachuma Formation in Agua Caliente Canyon for this study (Figure 6a) shows a single component of remanent magnetization that has a similar in situ direction to the present-day

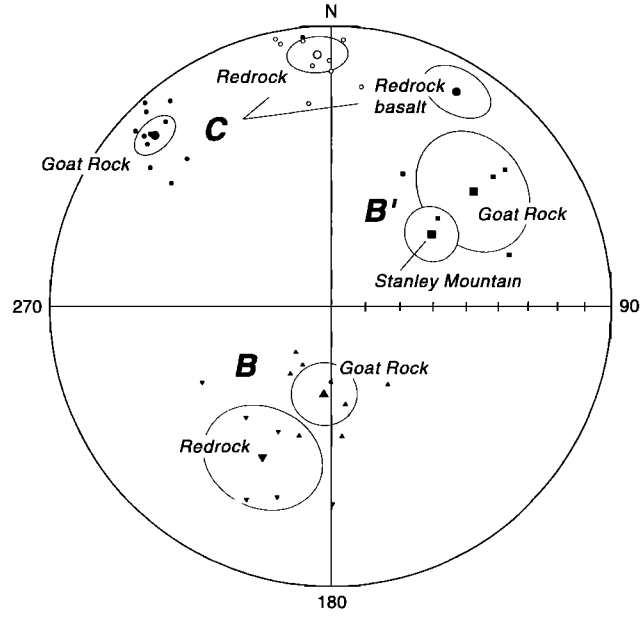


Figure 8. Equal-area plot of mean in situ directions for the *B* and *B'* components (squares and triangles, respectively), and corrected directions for the *C* components from Lower Jurassic chert samples taken near Goat Rock and in Redrock Canyon (circles), and from Lower Jurassic pillow basalts in Redrock Canyon (D. Smith et al., personal communication, 1993) and at Stanley Mountain [*McWilliams and Howell*, 1982]. Smaller symbols indicate sample/bedding layer means (Table 1). Component *C* exhibits near horizontal (equatorial paleolatitudes), and components *B* and *B'* might indicate middle Tertiary and/or younger remagnetizations and clockwise vertical-axis rotations. Solid symbols indicate lower hemisphere points; open symbols indicate upper hemisphere points.

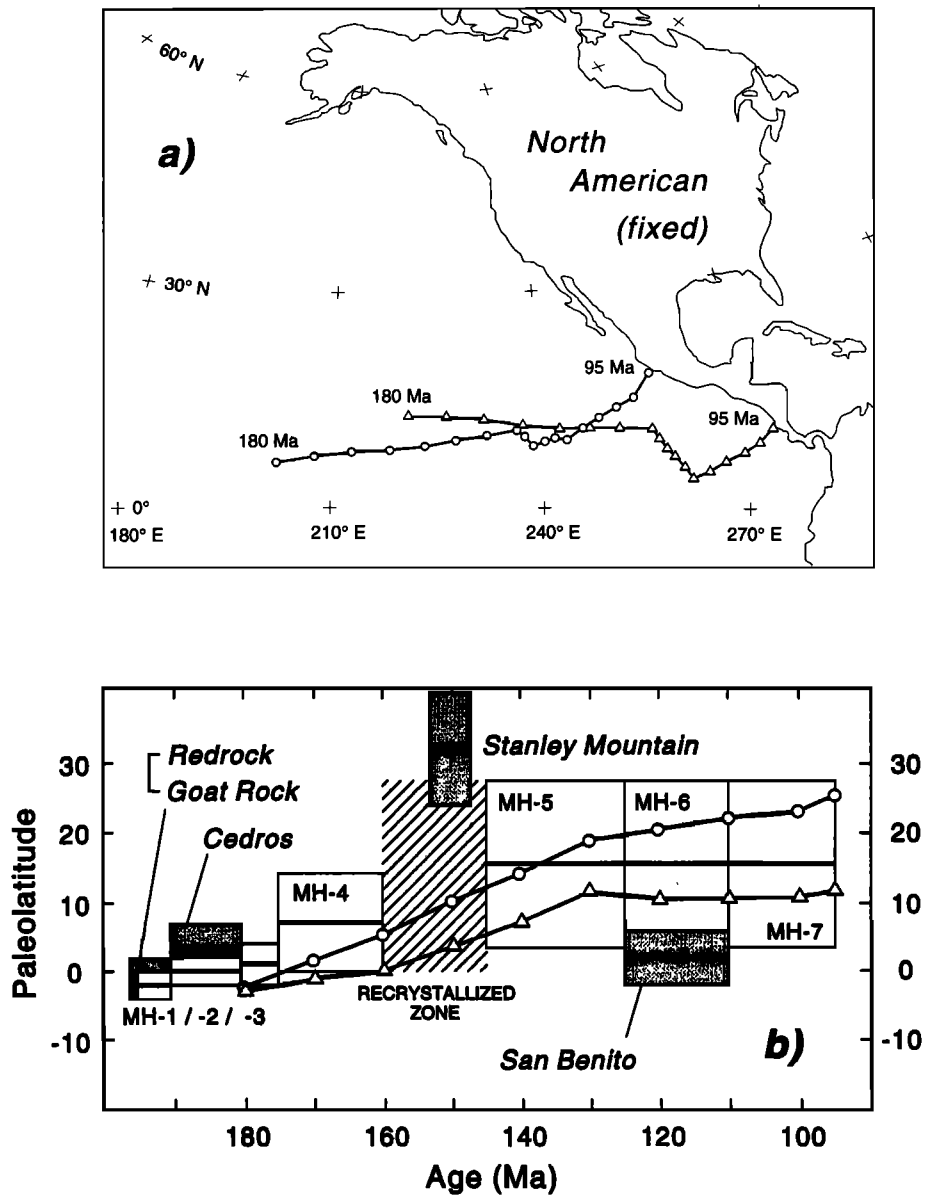


Figure 9. (a) Trajectories of Farallon plate motion in fixed North America coordinates determined by backward modeling between 95 Ma arrival times and 180 Ma from docking points (southern Mexico, 18°N; and Costa Rica, 10°N) selected along the present-day coastline [after *Debiche et al.*, 1987]. Symbols are plotted at 5-m.y. intervals along trajectories based on stage poles of *Engebretson et al.* [1985]. (b) Plot of paleolatitude versus time for Californian and Mexican chert sequences. Clear boxes labeled MH-1 through MH-7 (Marin Headlands biozones of *Murchey* [1984]) represent mean paleolatitudes (thick line) and 95% confidence limits for Franciscan chert sections in northern California [after *Hagstrum and Murchey*, 1993]. Similar shaded boxes have been plotted for combined data for the Goat Rock and Redrock Canyon cherts, for Jurassic and Cretaceous cherts from Cedros Island and the San Benito Islands, and for the Stanley Mountain tuffaceous chert [*Hagstrum et al.*, 1995]. Connected circles and triangles (at 5-m.y. intervals) represent trajectories of points on the Farallon plate shown in Figure 9a relative to the spin axis.

field and the *A* components found in the Goat Rock and Redrock cherts (Figure 7 and Table 2). Thermal demagnetizations of the Cachuma Formation samples also isolate the *A* component (Figure 6b) but often show complex behavior at the highest demagnetization steps suggesting the presence of a second high-temperature component of magnetization. Two samples (Figure 6c) contain a higher-coercivity component similar to the *B* component observed in the chert samples, but these directions are too few for a reliable mean.

The Upper Cretaceous sedimentary rocks at Figueroa Mountain [*McWilliams and Howell*, 1982] have a similar characteristic magnetization to that of the Cachuma Formation and might be a recent overprint as well. The mean characteristic direction for the Figueroa Mountain samples has steeper inclinations (Figure 7) which might be influenced by remnants of a *B* and/or a primary component of magnetization. A 6° ± 5° paleolatitude of deposition for the Figueroa Mountain rocks calculated by *McWilliams and Howell* [1982] therefore is suspect.

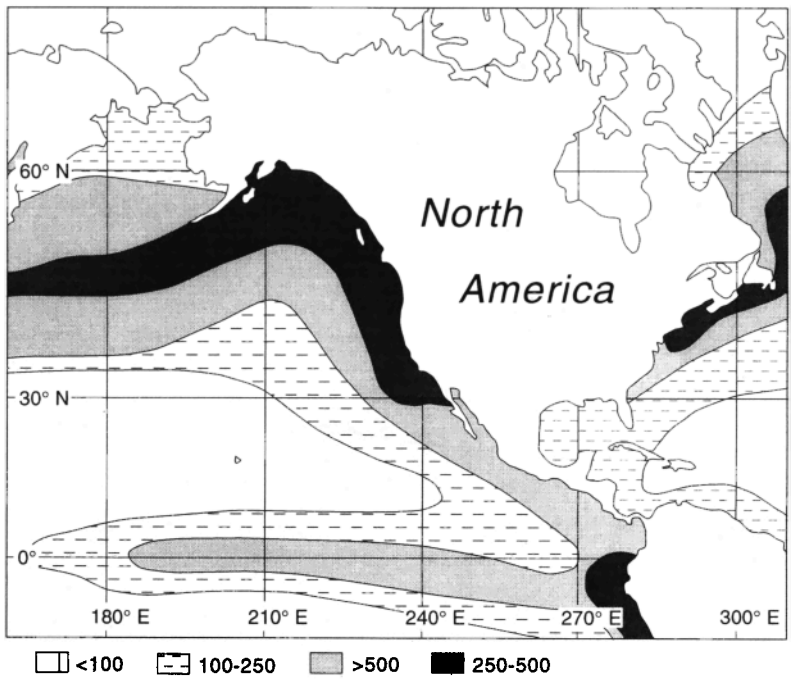


Figure 10. Annual production of silica by marine organisms in the eastern Pacific Ocean [*Lisitzin*, 1972]. Values in g m^{-2} yr^{-1}. These data, sedimentation rates, and the available paleomagnetic data indicate that deep-water Franciscan-type cherts were deposited within the ancient equatorial zone of high biologic productivity [*Hagstrum and Murchey*, 1993; *Hagstrum and Sedlock*, 1992; this study]. Tuffaceous chert above the Coast Range ophiolite at Stanley Mountain are inferred to have been deposited within a productivity high near the western margin of North America near ~30°N [*Hagstrum et al.*, 1995].

Blocking temperatures between 100° and 450°C and in situ directions near the recent axial dipole field for the Cachuma Formation in Agua Caliente Canyon are similar to results for other Upper Cretaceous forearc turbidite sequences in southern California and Baja California [*Hagstrum et al.*, 1985; *Fry et al.*, 1985; *Morris et al.*, 1986; *Filmer and Kirschvink*, 1989]. Although the characteristic magnetizations in most of these studies pass fold tests, the age of folding is generally poorly constrained and could be relatively recent. Several of the characteristic magnetizations in these studies also pass reversal tests which support but do not necessarily prove, early acquisition of the characteristic magnetization. Thus our result for the Cachuma Formation turbidites indicates that well-constrained field stablility tests are needed in making reliable interpretations of the characteristic magnetizations within these rocks and that in some cases secondary magnetizations might be incorrectly interpreted as primary magnetizations.

Conclusions

Red radiolarian chert exposed in the San Rafael Mountains of southern California apparently retains a primary magnetization (*C* component) indicating deposition of these sediments within the equatorial zone of high biologic productivity (Table 1 and Figure 10). This result is identical to that for coeval cherts within the Franciscan Complex of northern California (Figure 9), implying that these rocks have similar origins on the Farallon plate [*Hagstrum and Murchey*, 1993]. Accumulation of chert sequences was most likely sustained by eastward movement of the Farallon plate relative to North America beneath the equatorial productivity high. The chert sequence on Cedros Island in Baja California also probably shared this same depositional and translational history on the Farallon plate.

In contrast to other paleogeographic models of radiolarian chert deposition [e.g., *Pessagno and Blome*, 1986], we propose that chert sections now exposed along the Pacific rim of North America were initially deposited within zones of high biologic productivity at high paleolatitudes, equatorial paleolatitudes, and along the western continental margin (Figure 10). The association of Mesozoic chert deposition with siliceous planktonic productivity is supported by current silica sedimentation rates, and the margin of the equatorial productivity high is clearly shown by data for the chert section at the Marine Headlands: sedimentation rates fall off sharply upsection as the paleomagnetically-determined paleolatitude increases above ~7°N paleolatitude [*Hagstrum and Murchey*, 1993]. Deep-water Franciscan-type cherts therefore are interpreted to have been associated with the equatorial productivity zone, and tuffaceous cherts, such as those at Stanley Mountain in southern California [*Hagstrum et al.*, 1995], are interpreted to have been associated with the near-shore zone.

Mid-Cretaceous turbidites conformably overlie the radiolarian cherts in the Marin Headlands and on Cedros Island, indicating approach and accretion of the cherts to the continental margin at about this time. Paleomagnetic data from the uppermost part of the chert sequence in the Marin Headlands show that accretion occurred near 15° ± 12°N paleolatitude. The southern Mexican margin is a probable location because accretionary rocks have been tectonically removed from this region [*Karig et al.*, 1978; *Hagstrum et al.*, 1985]; farther south, such rocks are still present in Costa Rica and Panama. Thus chert sequences accreted from the Farallon plate in southern

Mexico were subsequently displaced northward and dispersed along the western margin of California and northern Mexico. likely by margin-parallel strike-slip faulting associated with poleward movements of the Kula, Farallon, and Pacific plates [*Hagstrum and Murchey*, 1993].

Tuffaceous cherts deposited in near-shore environments are generally more likely to be parautochthonous than farther-traveled equatorial cherts, although paleomagnetic data for Mid-Pennsylvanian to Lower Permian argillaceous chert blocks within the Slide Mountain terrane of British Columbia indicate deposition near 9° ± 3°N and 2° ± 2°N paleolatitude and a net poleward translation of 20° ± 4° [*Richards et al.*, 1993]. Because the North American plate was much farther south in early Mesozoic time, accretion paleolatitudes for equatorial chert sequences at that time could be similar to their paleolatitudes of deposition. Chert sequences accreted early on might also be less vulnerable to changes in plate motions that could detach and translate younger outboard terranes. Older terranes, however, tend to be more altered and thus less amenable to paleomagnetic investigations. Mesozoic subduction complexes at any given latitude along the American margin therefore likely contain a mixture of chert blocks with different depositional, accretion, and margin-parallel displacement histories.

The other components of magnetization observed in the Lower Jurassic cherts and Upper Cretaceous sedimentary rocks of this study imply that these rocks have complicated geologic histories; ages of these components are poorly constrained and thus interpretations of their origins are tentative. The *A* component found in the Lower Jurassic chert blocks and Upper Cretaceous Cachuma Formation is inferred to be related to recent uplift. Near complete remagnetization of the Cachuma sandstones and siltstones indicates that similar rocks in southern California and Baja California might also give unreliable results. The *B′* and *B* components are difficult to interpret because attitudes of the host rocks are unknown at the time of remagnetization. These components, however, are inferred to be overprint magnetizations associated with regional Miocene volcanism and subduction/accretion, respectively. Red chert so far is the sole rock type within the Franciscan Complex that retains a primary magnetization passing both fold and reversal tests, and consistent results from these deep-water cherts indicate that they have similar equatorial origins. More data, however, are needed from similar rocks along the margin to further test our model and determine the paleogeographic origins and displacement histories of these rocks.

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R. S. Bogar, Century West Engineering Corporation, 7950 Dublin Blvd., Suite 203, Dublin, CA 94568.

J. T. Hagstrum, U.S. Geological Survey, MS 937, 345 Middlefield Road, Menlo Park, CA 94025. (e-mail: jhag@mojave.wr.usgs.gov)

B. L. Murchey, U.S. Geological Survey, MS 915, 345 Middlefield Road, Menlo Park, CA 94025.

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